

# MIT 18.C06 - Fall 2026 - Calendar:

September 9<sup>th</sup>: Lecture 1 (Vectors, Matrices, Matrix-Vector Product)

September 11<sup>th</sup>: Lecture 2 (Row and Column Views of Linear Systems)

September 14<sup>th</sup>: Lecture 3 (Solving Linear Equations, REF and Backsubstitution)

September 14<sup>th</sup>: Pset 1 Out

September 16<sup>th</sup>: Lecture 4 (RREF, Matrix Multiplication and Counting)

September 18<sup>th</sup>: Lecture 5 (Properties of Matrix Multiplication and Inverses)

September 21<sup>st</sup>: Lecture 6 (Fundamental Operations with Matrices)

September 21<sup>st</sup>: Pset 1 Due

September 21<sup>st</sup>: Pset 2 Out

September 23<sup>rd</sup>: Lecture 7 (The Column Space and Nullspace)

September 25<sup>th</sup>: Lecture 8 (Linear Independence, Dimension and Bases)

September 28<sup>th</sup>: Lecture 9 (The Rank of a Matrix)

September 28<sup>th</sup>: Pset 2 Due

September 28<sup>th</sup>: Pset 3 Out

September 30<sup>th</sup>: Lecture 10 (Orthogonality and Gram-Schmidt)

October 2<sup>nd</sup>: Lecture 11 (Projections and Orthogonal Matrices)

October 5<sup>th</sup>: Lecture 12 (Review)

October 5<sup>th</sup>: Pset 3 Due

October 5<sup>th</sup>: Practice Exams Out

October 6<sup>th</sup>: Project 1 + Mini Pset Out

October 7<sup>th</sup>: Midterm 1 (Lectures 1-11)

October 9<sup>th</sup>: Lecture 13 (The Determinant and its Properties)

October 9<sup>th</sup>: Add date

October 12<sup>th</sup>: Indigenous Peoples' Day

October 13<sup>th</sup>: Monday schedule of classes to be held

October 13<sup>th</sup>: Lecture 14 (Existence of Matrix Inverse)

October 14<sup>th</sup>: Lecture 15 (Least Squares and Regularization)

October 16<sup>th</sup>: Lecture 16 (The Singular Value Decomposition)

October 16<sup>th</sup>: Pset 4 Out

October 19<sup>th</sup>: Lecture 17 (Low Rank Approximation)

October 19<sup>th</sup>: Project 1 + Mini Pset Due

October 21<sup>st</sup>: Lecture 18 (Eigenvalues and Eigenvectors)

October 23<sup>rd</sup>: Lecture 19 (The Eigendecomposition and Multiplicity)

October 23<sup>rd</sup>: Pset 4 Due

October 23<sup>rd</sup>: Pset 5 Out

October 26<sup>th</sup>: Lecture 20 (Principal Component Analysis)

October 28<sup>th</sup>: Lecture 21 (Word Embeddings and Biases in Data)

October 30<sup>th</sup>: Lecture 22 (The Power Method)

October 30<sup>th</sup>: Pset 5 Due

November 2<sup>nd</sup>: Lecture 23 (Review)

November 4<sup>th</sup>: Midterm 2 (Lectures 1-22)

November 6<sup>th</sup>: Lecture 23 (Symmetric Matrices)

November 6<sup>th</sup>: Pset 6 Out

November 9<sup>th</sup>: Lecture 24 (Backwards Connections of SVD)

November 11<sup>th</sup>: Veterans Day

November 13<sup>th</sup>: Lecture 26 (Nonnegative Matrices and Perron-Frobenius)

November 16<sup>th</sup>: Lecture 27 (Quadratic Programming)

November 16<sup>th</sup>: Pset 6 Due

November 16<sup>th</sup>: Project 2 + Pset 7 Out

November 18<sup>th</sup>: Lecture 28 (Local and Global Optimality Conditions)

November 18<sup>th</sup>: Drop date

November 20<sup>th</sup>: Lecture 29 (QP with Inequality Constraints)

November 23<sup>rd</sup>: Lecture 30 (Convex Sets and Convex Functions)

November 25<sup>th</sup>: Lecture 31 (Optimality and Gradient Descent)

November 25<sup>th</sup>: Project 2 + Pset 7 Due

November 26<sup>th</sup>: Thanksgiving

November 27<sup>th</sup>: Institute holiday

November 30<sup>th</sup>: Lecture 32 (Review)

December 2<sup>nd</sup>: Midterm 3 (Lectures 1-31)

December 4<sup>th</sup>: Lecture 33 (Understanding Gradient Descent)

December 4<sup>th</sup>: Last day any assignment can be due

December 7<sup>th</sup>: Lecture 33 (Understanding Gradient Descent, continued)

December 9<sup>th</sup>: Lecture 34 (Stochastic Gradient Descent)

December 10<sup>th</sup>: Last day of classes